Details on Cisco NX-OS and IOS-XE for AI-Architecture (Cisco-EN CLI Mapping)

Verification with Google NotebookLLM (April 1, 2026 6am Pacific)

- If you share your Google email I can activate you on my notebook to view the research and collaborate in real time
- Khursheed's NotebookLLM_CLI-Mappings (CISCO-EN)

Commented [KK1]: @James, @James, this is the first assignment where collaboration is key. Google NotebookLLM is a good starting point for deep research prior to any AI-work with ClaudeCode or architecture. If you have a google gmail account I can add you to the collaboration list and we can jointly develop the 'brains' ?

Question from ,

- In your notes I did see NexOS accounted for and later in the documentation, IOS. Can you provide the versions (code train) as this will impact the success or challenge for the agent and output ....

Response from Khursheed,

- We have the most updated reference that Cisco publishes online that the agent will query.
    - - IOS-XE: 17.18.1 documented on cisco.com; 17.12.x (Dublin) most active; 17.15 = Extended-Support
    - - NX-OS: 10.5(5)M released March 16, 2026; 10.6(x) upgrade guides live on cisco.com
- Details below with actual links for references.

Response Details

Cisco **NX-OS** and **IOS-XE** represent two distinct architectural philosophies within the Cisco ecosystem, with **NX-OS**tailored for high-density data center environments and **IOS-XE** acting as a unified operating system for campus access, routing, and wireless portfolios.

1. Cisco NX-OS: Model-Driven Data Center Architecture

NX-OS is built upon a **Model-Driven Programmability (MDP)** framework, allowing for granular automation and monitoring of the device state.

- **Centralized DME Database:** The foundation of NX-OS is a centralized **Data Management Engine (DME)**database. All configuration and operational states are stored here, ensuring that the CLI, SNMP, and programmatic APIs (YANG models) always provide a **consistent view** of the switch without discrepancy.
- **Modular Feature-Based Logic:** Unlike traditional IOS, NX-OS is highly modular. You must **explicitly "enable" features** (e.g., `feature ospf`, `feature lacp`, or `feature nxapi`) before the associated configuration commands become available in the CLI.
- **Programmability & On-Box Shells:** NX-OS supports standards-based agents like **NETCONF, RESTCONF, and gRPC** using YANG data modeling. It also features a **Guestshell**, which is a secure Linux container where administrators can run Python scripts or third-party applications directly on the hardware.

- **Data Center Focus:** The OS is optimized for 400G and 800G hardware, utilizing tools like **ethanalyzer** (built on Wireshark) for deep local packet capture and analysis on the Nexus 9000 series.

2. Cisco IOS-XE: Unified Campus & Enterprise OS

IOS-XE is a Linux-based evolution of classic IOS that now offers a **single software image** across the Catalyst switching, routing, and wireless portfolio for easier lifecycle management.

- **Hierarchical Command Modes:** The user interface is strictly divided into functional modes, including **User EXEC**, **Privileged EXEC**, **Global Configuration**, and specific sub-modes like **Interface Configuration**.
- **Hardware-Dictated CLI:** Because Catalyst 9000 hardware uses powerful, programmable **UADP ASICs**, the CLI is designed to handle complex, multi-stage lookups. This results in **deep sub-modes** (Global > Interface > Service-Policy) to manage how the hardware processes a single packet.
- **Transition to Install Mode:** Cisco is moving away from "Bundle Boot" in favor of **"Install Mode"** (specifically on the 17.15 train and later), which provides better performance and granular patching capabilities.
- **System Sanitization:** Modern IOS-XE supports the `factory-reset` command to completely clear local state, whereas older versions relied primarily on `write erase`.

3. Key Operational & Command Differences

While Cisco maintains high consistency between these OSs, there are critical syntax and logic variations:

Function	Cisco IOS-XE (Catalyst)	Cisco NX-OS (Nexus)
Prerequisites	Direct configuration allowed.	Must issue **`feature [name]`** first.
Negation	Uses **`no [command]`**.	Uses **`no [command]`**.
Interface Brief	`show ip interface brief`.	**`show interface brief`**.
Saving Config	`copy run start` or `write memory`.	**`copy running-config startup-config`**.
Packet Tracing	`show platform hardware fed ...`.	**`ethanalyzer local interface [id]`**.
Programmability	Focus on **Install Mode** and YANG.	Focus on **DME**, YANG, and **Guestshell**.

4. Contextual Integration (The "Rosetta Stone")

In the broader context of your **RAN-enabled AI-agent**, these OSs serve as the "source of truth" for hardware-specific instructions. While the industry is moving toward **Service-Abstraction** (orchestrating functions like "Onboarding" or "Segmentation"), the agent must be calibrated to recognize that **NX-OS requires "Feature" activation** before it can implement the same functional intent that **IOS-XE** might accept natively.

Details on Cisco NX-OS and IOS-XE for AI-Architecture (Cisco-EN CLI Mapping)

Based on the provided sources, which include documentation updated as recently as March 2026, the latest versions of Cisco's primary network operating systems are as follows:

Cisco IOS-XE

- **Latest Version: 17.18.1** is explicitly cited in the command references.
- **Release Lifecycle:** The **17.15 train** is currently considered the primary **"Long-Lived" release**.
- **Strategic Direction:** Cisco has announced that the traditional "Bundle Boot" mode will be discontinued starting with **Release 27.01**, necessitating a transition to "Install Mode" for enhanced performance and more granular patching.

Cisco NX-OS

- **Latest Versions:** The **10.6(x)** and **10.7(x)** releases are the current standard for modern data center environments, specifically optimized for high-density 400G and 800G hardware.
- **Specific Release:** Documentation from March 2026 references **Release 10.6(2)F**, which was published in December 2025.
- **Architecture:** NX-OS continues to rely on its centralized **Data Management Engine (DME) database** to ensure state consistency across CLI, SNMP, and programmatic interfaces (YANG).

These versions represent the "state of the art" for the Catalyst and Nexus hardware lines as of the first quarter of 2026.

Details on Cisco NX-OS and IOS-XE for AI-Architecture (Cisco-EN CLI Mapping)

Updated on AI-Architecture this morning, April 1, 2026 6:30am Pacific

What Was Updated Today

bin_onboard.md
- `factory-reset all` added as the 17.15+ replacement for `write erase`
- Install Mode / Bundle Boot deprecation (27.01) noted
- New Theme 7: Version reference table for all 6 OS platforms with caveat rule

bin_sys_info.md
- New Guestshell section: `feature guestshell` → `guestshell enable` → Linux container on-box; NX-OS exclusive; no IOS-XE equivalent

system_master.md (Rule 1)
- Added the DME "why" — `feature` unlocks a sub-tree, not just a CLI switch
- Added the **"Rosetta Stone" rule** — NX-OS needs feature activation for what IOS-XE accepts natively
- Added `feature guestshell` to the prereq list

Web Search Confirms (cisco.com, April 2026)

- **IOS-XE:** 17.18.1 documented on cisco.com; 17.12.x (Dublin) most active; 17.15 = Extended-Support
- **NX-OS:** 10.5(5)M released March 16, 2026; 10.6(x) upgrade guides live on cisco.com

NotebookLM data is accurate and web-verified.

Sources:
- https://www.cisco.com/c/en/us/support/ios-nx-os-software/ios-xe-17-18-1/model.html
- https://www.cisco.com/c/en/us/td/docs/switches/datacenter/nexus9000/sw/recommended_release/b_Minimum_and_Recommended_Cisco_NX-OS_Releases_for_Cisco_Nexus_9000_Series_Switches.html
- https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst-9000/collections/catalyst-9000-release-notes.html

- https://www.cisco.com/c/en/us/td/docs/dcn/nx-os/nexus9000/106x/upgrade/cisco-nexus-9000-series-nx-os-software-upgrade-and-downgrade-guide-106x/m-upgrade-or-downgrade-the-cisco-nexus-9000-series-nx-os-software.html